

第十七讲：数学期望（续）与方差

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- 若 $X(\omega) = \sum_{i=1}^n a_i I_{A_i}(\omega)$, $A_1, \dots, A_n \in \mathcal{F}$, 即 X 为简单随机变量时, 定义

$$\int_{\Omega} X(\omega) P(d\omega) := \sum_{i=1}^n a_i P(A_i)$$

► 若 $X(\omega)$ 为非负随机变量, 则由之前随机变量的结构易知:

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$$0 \leq X_n(\omega) \leq X_{n+1}(\omega) \leq X(\omega) \quad P\text{-a.s.}$$

且 $X_n(\omega)$ 为一简单函数, $X_n(\omega) = \sum_{i=1}^n a_{i,n} I_{A_{i,n}}(\omega)$

$$\lim_{n \rightarrow \infty} \int_{\Omega} X_n(\omega) P(d\omega) = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_{i,n} P(A_{i,n}) = \int_{\Omega} X(\omega) P(d\omega)$$



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若 $X(\omega)$ 为任一实数, 则令 $X_n(\omega) = X(\omega) \wedge n$



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1. 若 c 为常数, 则 $E(c) = c$;
2. 若 $X \geq 0$, 则 $E(X) \geq 0$, 且 $E(X) = 0$ 当且仅当 $X = 0$ 几乎处处成立;
3. 对任意的常数 a , 有 $E(aX) = aE(X)$;
4. 对任意两个函数 $g_1(x), g_2(x)$ 有

$$E(g_1(X) \pm g_2(X)) = E(g_1(X)) \pm E(g_2(X))$$

5. 对任意两个随机变量 X, Y , 有 $E(X+Y) = E(X) + E(Y)$;
6. 一般的, 对任意两个随机变量 X, Y 及任给两个常数 a, b , 有

$$E(aX + bY) = aE(X) + bE(Y);$$

7. 更一般的有

$$E\left(\sum_{i=1}^n a_i X_i + b\right) = \sum_{i=1}^n a_i E X_i + b$$



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定理 0.1 设 X 为非负随机变量. 则 $EX \geq 0$, 且 $EX = 0$ 当且仅当 $P(X = 0) = 1$, 即 $X = 0, a.s.$

证明:

► 因 X 是非负随机变量, 当 $x < 0$ 有 $F_X(x) = 0$.

► 由斯蒂尔切斯积分的定义和性质易知

$$EX = \int_0^{\infty} x dF_X(x) \geq 0.$$

► 若 $X = 0, a.s.$, 则 $F_X(x) = 1_{\{x \geq 0\}}$. 故 $E(X) = \int_{-\infty}^{\infty} x dF_X(x) = 0$.

► 反之, 若 $EX = 0$, 则对任意的 $a > 0$ 有

$$0 = \int_0^{\infty} x dF_X(x) \geq \int_a^{\infty} x dF_X(x) \geq a \int_a^{\infty} dF_X(x) = a [1 - F_X(a)]$$

► 故 $F_X(a) = 1$, 再由右连续性得 $F_X(0) = 1$, 从而 $X = 0, a.s.$

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- ▶ 若 $X = 0, a.s.$, 则 $F_X(x) = 1_{\{x \geq 0\}}$. 故 $E(X) = \int_{-\infty}^{\infty} x dF_X(x) = 0$.
- ▶ 反之, 若 $EX = 0$, 则对任意的 $a > 0$ 有

$$0 = \int_0^{\infty} x dF_X(x) \geq \int_a^{\infty} x dF_X(x) \geq a \int_a^{\infty} dF_X(x) = a [1 - F_X(a)]$$

- ▶ 故 $F_X(a) = 1$, 再由右连续性得 $F_X(0) = 1$, 从而 $X = 0, a.s.$



定理 0.1 设 X 为非负随机变量. 则 $EX \geq 0$, 且 $EX = 0$ 当且仅当 $P(X=0) = 1$, 即 $X=0, a.s..$

证明:

- ▶ 因 X 是非负随机变量, 当 $x < 0$ 有 $F_X(x) = 0$.
- ▶ 由斯蒂尔切斯积分的定义和性质易知

$$EX = \int_0^{\infty} x dF_X(x) \geq 0.$$

- ▶ 若 $X=0, a.s.$, 则 $F_X(x) = 1_{\{x \geq 0\}}$. 故 $E(X) = \int_{-\infty}^{\infty} x dF_X(x) = 0$.
- ▶ 反之, 若 $EX = 0$, 则对任意的 $a > 0$ 有

$$0 = \int_0^{\infty} x dF_X(x) \geq \int_a^{\infty} x dF_X(x) \geq a \int_a^{\infty} dF_X(x) = a [1 - F_X(a)]$$

- ▶ 故 $F_X(a) = 1$, 再由右连续性得 $F_X(0) = 1$, 从而 $X=0, a.s..$



定理 0.2 给定随机变量 X ，令 $X^+ = 0 \vee X$ 和 $X^- = 0 \vee (-X)$ ，则

$$EX^+ = \int_0^{\infty} x dF_X(x) = \int_0^{\infty} [1 - F_X(t)] dt$$
$$EX^- = - \int_{-\infty}^0 x dF_X(x) = \int_{-\infty}^0 F_X(t) dt$$



证明:

- ▶ 因 X^+, X^- 为非负随机变量, 故
 - ▶ 对 $x < 0$ 有 $F_{X^+}(x) = F_{X^-}(x) = 0$.
 - ▶ 对 $x \geq 0$, 有

$$F_{X^+}(x) = F_X(x), \quad F_{X^-}(x) = P(X \geq -x) = 1 - P(X < -x).$$

▶ 从而

$$\begin{aligned} EX^+ &= \int_0^\infty x dF_X(x) = \int_0^\infty dF_X(x) \int_0^x dt \\ &= \int_0^\infty dt \int_t^\infty dF_X(x) = \int_0^\infty [1 - F_X(t)] dt \\ EX^- &= \int_0^\infty [1 - F_{X^-}(x)] dx = \int_{-\infty}^0 [1 - F_X(-x)] dx \\ &= \int_{-\infty}^0 P(X < x) dx = \int_{-\infty}^0 F_X(x) dx \end{aligned}$$

▶ 最后一个等号是因为改变可数个点上的函数值



证明:

- ▶ 因 X^+, X^- 为非负随机变量, 故
 - ▶ 对 $x < 0$ 有 $F_{X^+}(x) = F_{X^-}(x) = 0$.
 - ▶ 对 $x \geq 0$, 有

$$F_{X^+}(x) = F_X(x), \quad F_{X^-}(x) = P(X \geq -x) = 1 - P(X < -x).$$

▶ 从而

$$\begin{aligned} EX^+ &= \int_0^\infty x dF_X(x) = \int_0^\infty dF_X(x) \int_0^x dt \\ &= \int_0^\infty dt \int_t^\infty dF_X(x) = \int_0^\infty [1 - F_X(t)] dt \\ EX^- &= \int_0^\infty [1 - F_{X^-}(x)] dx = \int_{-\infty}^0 [1 - F_X(-x)] dx \\ &= \int_{-\infty}^0 P(X < x) dx = \int_{-\infty}^0 F_X(x) dx \end{aligned}$$

▶ 最后一个等号是因为改变可数个点上的函数值不影响黎曼积分值

证明:

- ▶ 因 X^+, X^- 为非负随机变量, 故
 - ▶ 对 $x < 0$ 有 $F_{X^+}(x) = F_{X^-}(x) = 0$.
 - ▶ 对 $x \geq 0$, 有

$$F_{X^+}(x) = F_X(x), \quad F_{X^-}(x) = P(X \geq -x) = 1 - P(X < -x).$$

▶ 从而

$$\begin{aligned} EX^+ &= \int_0^\infty x dF_X(x) = \int_0^\infty dF_X(x) \int_0^x dt \\ &= \int_0^\infty dt \int_t^\infty dF_X(x) = \int_0^\infty [1 - F_X(t)] dt \\ EX^- &= \int_0^\infty [1 - F_{X^-}(x)] dx = \int_{-\infty}^0 [1 - F_{X^-}(-x)] dx \\ &= \int_{-\infty}^0 P(X < x) dx = \int_{-\infty}^0 F_X(x) dx \end{aligned}$$

▶ 最后一个等号是因为改变可数个点上的函数值不影响黎曼积分值。

证明:

- ▶ 因 X^+, X^- 为非负随机变量, 故
 - ▶ 对 $x < 0$ 有 $F_{X^+}(x) = F_{X^-}(x) = 0$.
 - ▶ 对 $x \geq 0$, 有

$$F_{X^+}(x) = F_X(x), \quad F_{X^-}(x) = P(X \geq -x) = 1 - P(X < -x).$$

▶ 从而

$$\begin{aligned} EX^+ &= \int_0^\infty x dF_X(x) = \int_0^\infty dF_X(x) \int_0^x dt \\ &= \int_0^\infty dt \int_t^\infty dF_X(x) = \int_0^\infty [1 - F_X(t)] dt \\ EX^- &= \int_0^\infty [1 - F_{X^-}(x)] dx = \int_{-\infty}^0 [1 - F_X(-x)] dx \\ &= \int_{-\infty}^0 P(X < x) dx = \int_{-\infty}^0 F_X(x) dx \end{aligned}$$

▶ 最后一个等号是因为改变可数个点上的函数值不影响黎曼积分值。

证明:

- ▶ 因 X^+, X^- 为非负随机变量, 故
 - ▶ 对 $x < 0$ 有 $F_{X^+}(x) = F_{X^-}(x) = 0$.
 - ▶ 对 $x \geq 0$, 有

$$F_{X^+}(x) = F_X(x), \quad F_{X^-}(x) = P(X \geq -x) = 1 - P(X < -x).$$

▶ 从而

$$\begin{aligned} EX^+ &= \int_0^\infty x dF_X(x) = \int_0^\infty dF_X(x) \int_0^x dt \\ &= \int_0^\infty dt \int_t^\infty dF_X(x) = \int_0^\infty [1 - F_X(t)] dt \\ EX^- &= \int_0^\infty [1 - F_{X^-}(x)] dx = \int_{-\infty}^0 [1 - F_X(-x)] dx \\ &= \int_{-\infty}^0 P(X < x) dx = \int_{-\infty}^0 F_X(x) dx \end{aligned}$$

▶ 最后一个等号是因为改变可数个点上的函数值不影响黎曼积分值。

定理 0.3 若随机变量 X 的数学期望存在, 则有

$$EX = \int_0^{\infty} P(X > t) dt - \int_{-\infty}^0 P(X \leq t) dt$$
$$E|X| = \int_0^{\infty} P(X > t) dt + \int_{-\infty}^0 P(X \leq t) dt$$

例 0.1 设 X 为取值于数集 $\{0, 1, 2, \dots\}$ 的离散型随机变量. 则

$$\begin{aligned} EX &= \int_0^{\infty} P(X > t) dt = \sum_{n=1}^{\infty} \int_{n-1}^n P(X > t) dt \\ &= \sum_{n=1}^{\infty} \int_{n-1}^n P(X \geq n) dt = \sum_{n=1}^{\infty} P(X \geq n) \end{aligned}$$

定理 0.3 若随机变量 X 的数学期望存在, 则有

$$EX = \int_0^{\infty} P(X > t) dt - \int_{-\infty}^0 P(X \leq t) dt$$
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定理 0.4 设随机变量 X 有分布函数 F_X ，而 f 为 $\mathbb{R}$ 上的可测函数，则随机变量 $f(X)$ 的数学期望存在当且仅当 f 的 F_X 积分存在，且此时

$$Ef(X) = \int_{-\infty}^{\infty} f(x) dF_X(x)$$

证明： 由定理0.3知：

$$\begin{aligned} Ef(X) &= \int_0^{\infty} P(f(X) > t) dt - \int_{-\infty}^0 P(f(X) \leq t) dt \\ &= \int_0^{\infty} dt \int_{-\infty}^{\infty} 1_{\{f(x) > t\}} dF_X(x) - \int_{-\infty}^0 dt \int_{-\infty}^{\infty} 1_{\{f(x) \leq t\}} dF_X(x) \\ &= \int_{-\infty}^{\infty} dF_X(x) \int_0^{\infty} 1_{\{f(x) > t\}} dt - \int_{-\infty}^{\infty} dF_X(x) \int_{-\infty}^0 1_{\{f(x) \leq t\}} dt \\ &= \int_{-\infty}^{\infty} f^+(x) dF_X(x) - \int_{-\infty}^{\infty} f^-(x) dF_X(x) = \int_{-\infty}^{\infty} f(x) dF_X(x) \end{aligned}$$



定理 0.4 设随机变量 X 有分布函数 F_X ，而 f 为 $\mathbb{R}$ 上的可测函数，则随机变量 $f(X)$ 的数学期望存在当且仅当 f 的 F_X 积分存在，且此时

$$Ef(X) = \int_{-\infty}^{\infty} f(x) dF_X(x)$$

证明： 由定理0.3知：

$$\begin{aligned} Ef(X) &= \int_0^{\infty} P(f(X) > t) dt - \int_{-\infty}^0 P(f(X) \leq t) dt \\ &= \int_0^{\infty} dt \int_{-\infty}^{\infty} 1_{\{f(x) > t\}} dF_X(x) - \int_{-\infty}^0 dt \int_{-\infty}^{\infty} 1_{\{f(x) \leq t\}} dF_X(x) \\ &= \int_{-\infty}^{\infty} dF_X(x) \int_0^{\infty} 1_{\{f(x) > t\}} dt - \int_{-\infty}^{\infty} dF_X(x) \int_{-\infty}^0 1_{\{f(x) \leq t\}} dt \\ &= \int_{-\infty}^{\infty} f^+(x) dF_X(x) - \int_{-\infty}^{\infty} f^-(x) dF_X(x) = \int_{-\infty}^{\infty} f(x) dF_X(x) \end{aligned}$$



定理 0.5 若随机变量 X 的分布函数为 $F(x)$, 则 X 的某一函数 $g(X)$ 的期望为

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{+\infty} g(x) dF(x) \\ &= \begin{cases} \sum_i g(x_i) [F(x_i) - F(x_{i-1})] = \sum_i g(x_i) P(X = x_i), \\ \int_{-\infty}^{+\infty} g(x) p(x) dx. \end{cases} \end{aligned}$$



定理 0.5 若随机变量 X 的分布函数为 $F(x)$, 则 X 的某一函数 $g(X)$ 的期望为

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{+\infty} g(x) dF(x) \\ &= \begin{cases} \sum_i g(x_i) [F(x_i) - F(x_{i-1})] = \sum_i g(x_i) P(X = x_i), \\ \int_{-\infty}^{+\infty} g(x) p(x) dx. \end{cases} \end{aligned}$$



定理 0.6 设 X 为概率空间 $(\Omega, \mathcal{F}, P)$ 上的随机变量, $\mathbf{F}$ 为随机变量 X 的分布, 则对任意的 Borel 函数 $f(x)$ 有

$$\int_{\Omega} f(X(\omega))P(d\omega) = \int_{\mathbb{R}} f(x)\mathbf{F}(dx) \left(= \int_{\mathbb{R}} f(x)dF(x) \right).$$

特别的, 若 $f(x) = x$, 则有

$$E(X) = \int_{\Omega} X(\omega)P(d\omega) = \int_{\mathbb{R}} x\mathbf{F}(dx) = \int_{\mathbb{R}} xdF(x)$$



定理 0.6 设 X 为概率空间 $(\Omega, \mathcal{F}, P)$ 上的随机变量, $\mathbf{F}$ 为随机变量 X 的分布, 则对任意的 Borel 函数 $f(x)$ 有

$$\int_{\Omega} f(X(\omega))P(d\omega) = \int_{\mathbb{R}} f(x)\mathbf{F}(dx) \left(= \int_{\mathbb{R}} f(x)dF(x) \right).$$

特别的, 若 $f(x) = x$, 则有

$$E(X) = \int_{\Omega} X(\omega)P(d\omega) = \int_{\mathbb{R}} x\mathbf{F}(dx) = \int_{\mathbb{R}} xdF(x)$$



定理 0.7 若随机向量 $(X_1, \dots, X_n)$ 的分布函数为 $F(x_1, \dots, x_n)$, $g(x_1, \dots, x_n)$ 为 n 元 Borel 函数, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) dF(x_1, \dots, x_n).$$

特别的,

$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i dF(x_1, \dots, x_n) = \int_{-\infty}^{+\infty} x_i dF_i(x_i)$$

其中 $F_i(x_i)$ 为 X_i 的分布函数. 更进一步, 如果 $(X_1, \dots, X_n)$ 为连续型随机向量即具有联合分布密度 $p(x_1, \dots, x_n)$, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) p(x_1, \dots, x_n) dx_1 \cdots dx_n$$
$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i p(x_1, \dots, x_n) dx_1 \cdots dx_n = \int_{-\infty}^{+\infty} x_i p_{X_i}(x_i) dx_i$$



定理 0.7 若随机向量 $(X_1, \dots, X_n)$ 的分布函数为 $F(x_1, \dots, x_n)$, $g(x_1, \dots, x_n)$ 为 n 元 Borel 函数, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) dF(x_1, \dots, x_n).$$

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$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i dF(x_1, \dots, x_n) = \int_{-\infty}^{+\infty} x_i dF_i(x_i)$$

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$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) p(x_1, \dots, x_n) dx_1 \cdots dx_n$$

$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i p(x_1, \dots, x_n) dx_1 \cdots dx_n = \int_{-\infty}^{+\infty} x_i p_{X_i}(x_i) dx_i$$



定理 0.7 若随机向量 $(X_1, \dots, X_n)$ 的分布函数为 $F(x_1, \dots, x_n)$, $g(x_1, \dots, x_n)$ 为 n 元 Borel 函数, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) dF(x_1, \dots, x_n).$$

特别的,

$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i dF(x_1, \dots, x_n) = \int_{-\infty}^{+\infty} x_i dF_i(x_i)$$

其中 $F_i(x_i)$ 为 X_i 的分布函数. 更进一步, 如果 $(X_1, \dots, X_n)$ 为连续型随机向量即具有联合分布密度 $p(x_1, \dots, x_n)$, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) p(x_1, \dots, x_n) dx_1 \cdots dx_n$$
$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i p(x_1, \dots, x_n) dx_1 \cdots dx_n = \int_{-\infty}^{+\infty} x_i p_{X_i}(x_i) dx_i$$



定理 0.7 若随机向量 $(X_1, \dots, X_n)$ 的分布函数为 $F(x_1, \dots, x_n)$, $g(x_1, \dots, x_n)$ 为 n 元 Borel 函数, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) dF(x_1, \dots, x_n).$$

特别的,

$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i dF(x_1, \dots, x_n) = \int_{-\infty}^{+\infty} x_i dF_i(x_i)$$

其中 $F_i(x_i)$ 为 X_i 的分布函数. 更进一步, 如果 $(X_1, \dots, X_n)$ 为连续型随机向量即具有联合分布密度 $p(x_1, \dots, x_n)$, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) p(x_1, \dots, x_n) dx_1 \cdots dx_n$$

$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i p(x_1, \dots, x_n) dx_1 \cdots dx_n = \int_{-\infty}^{+\infty} x_i p_{X_i}(x_i) dx_i$$



定理 0.7 若随机向量 $(X_1, \dots, X_n)$ 的分布函数为 $F(x_1, \dots, x_n)$, $g(x_1, \dots, x_n)$ 为 n 元 Borel 函数, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) dF(x_1, \dots, x_n).$$

特别的,

$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i dF(x_1, \dots, x_n) = \int_{-\infty}^{+\infty} x_i dF_i(x_i)$$

其中 $F_i(x_i)$ 为 X_i 的分布函数. 更进一步, 如果 $(X_1, \dots, X_n)$ 为连续型随机向量即具有联合分布密度 $p(x_1, \dots, x_n)$, 则

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} g(x_1, \dots, x_n) p(x_1, \dots, x_n) dx_1 \cdots dx_n$$
$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i p(x_1, \dots, x_n) dx_1 \cdots dx_n = \int_{-\infty}^{+\infty} x_i p_{X_i}(x_i) dx_i$$



定义 0.3 假设随机向量 $(X_1, \dots, X_n)$ 的每个分量 X_i 的数学期望都存在, 则称

$$E(X) = (EX_1, EX_2, \dots, EX_n)$$

为随机向量 X 的数学期望向量, 简称为 X 的数学期望。这里

$$E(X_i) = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} x_i dF(x_1, \dots, x_n) = \int_{-\infty}^{+\infty} x_i dF_i(x_i)$$

这里 $F_i(x_i)$ 为 X_i 的分布函数.



定理 0.8 设 X, Y 为概率空间 $(\Omega, \mathcal{F}, P)$ 上两个相互独立的随机变量, 且 X, Y 均可积, 则乘积 XY 也可积, 且

$$E(XY) = E(X)E(Y)$$

定理 0.9 概率空间 $(\Omega, \mathcal{F}, P)$ 上两个随机变量 X, Y 相互独立的充要条件是, 对于使得 $f(X)$ 与 $g(Y)$ 均可积的任何 Borel 函数 f, g 均有

$$E[f(X)g(Y)] = E[f(X)]E[g(Y)].$$



定理 0.8 设 X, Y 为概率空间 $(\Omega, \mathcal{F}, P)$ 上两个相互独立的随机变量, 且 X, Y 均可积, 则乘积 XY 也可积, 且

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定理 0.9 概率空间 $(\Omega, \mathcal{F}, P)$ 上两个随机变量 X, Y 相互独立的充要条件是, 对于使得 $f(X)$ 与 $g(Y)$ 均可积的任何 Borel 函数 f, g 均有

$$E[f(X)g(Y)] = E[f(X)]E[g(Y)].$$



定理 0.10 (单调收敛定理) 若随机变量序列 $\{X_n\}$ 满足条件

$$0 \leq X_1(\omega) \leq X_2(\omega) \leq \cdots \leq X_n(\omega) \uparrow X(\omega), \omega \in \Omega,$$

则

$$\begin{aligned} \int_{\Omega} \lim_{n \rightarrow \infty} X_n(\omega) P(d\omega) &= \lim_{n \rightarrow \infty} \int_{\Omega} X_n(\omega) P(d\omega), \\ \text{i.e. } E(\lim_{n \rightarrow \infty} X_n) &= \lim_{n \rightarrow \infty} E(X_n). \end{aligned}$$



- 设 $\{X_{nk}\}_{k \geq 1}$ 为定义 $E(X_n)$ 的简单随机变量列, 且满足

$$0 \leq X_{11} \leq X_{12} \leq \cdots \leq X_{1k} \uparrow X_1$$

$$0 \leq X_{21} \leq X_{22} \leq \cdots \leq X_{2k} \uparrow X_2$$

$$\vdots$$

$$0 \leq X_{n1} \leq X_{n2} \leq \cdots \leq X_{nk} \uparrow X_n$$

$$\lim_{k \rightarrow \infty} E(X_{nk}) = E(X_n)$$

- 令 $\tilde{X}_k = \max_{1 \leq i \leq k} X_{ik}$, 则 $\tilde{X}_k$ 为非降简单随机变量序列,

$$X_{nk} \leq \tilde{X}_k \leq X_k, \quad E(X_{nk}) \leq E(\tilde{X}_k) \leq E(X_k)$$



► 下证 $\lim_{k \rightarrow \infty} \tilde{X}_k = X$, $\lim_{k \rightarrow \infty} E(\tilde{X}_k) = \lim_{n \rightarrow \infty} E(X_n)$

► 固定 n , 令 $k \rightarrow +\infty$ 可得

$$X_n \leq \lim_{k \rightarrow +\infty} \tilde{X}_k \leq X, \quad E(X_n) \leq \lim_{k \rightarrow +\infty} E(\tilde{X}_k) \leq \lim_{k \rightarrow +\infty} E(X_k)$$

► 再令 $n \rightarrow +\infty$ 可得

$$\lim_{k \rightarrow \infty} \tilde{X}_k = X, \quad \lim_{k \rightarrow \infty} E(\tilde{X}_k) = \lim_{n \rightarrow \infty} E(X_n)$$

► 结合 $E(\lim_{n \rightarrow \infty} X_n) = E(X) := \lim_{k \rightarrow \infty} E(\tilde{X}_k) = \lim_{n \rightarrow \infty} E(X_n)$ 可知定理得证.



定理 0.11 (Fatou 引理) 若 $\{X_n\}$ 是一随机变量序列,

► 如果存在可积随机变量 $\underline{Y}$ 使得 $X_n \geq \underline{Y}$, 则有

$$\begin{aligned} \int_{\Omega} \liminf_{n \rightarrow \infty} X_n(\omega) P(d\omega) &\leq \liminf_{n \rightarrow \infty} \int_{\Omega} X_n(\omega) P(d\omega), \\ \text{i.e. } E(\liminf_{n \rightarrow \infty} X_n) &\leq \liminf_{n \rightarrow \infty} E(X_n). \end{aligned}$$

► 如果存在可积随机变量 $\bar{Y}$ 使得 $X_n \leq \bar{Y}$, 则有

$$\begin{aligned} \int_{\Omega} \limsup_{n \rightarrow \infty} X_n(\omega) P(d\omega) &\geq \limsup_{n \rightarrow \infty} \int_{\Omega} X_n(\omega) P(d\omega), \\ \text{i.e. } E(\limsup_{n \rightarrow \infty} X_n) &\geq \limsup_{n \rightarrow \infty} E(X_n). \end{aligned}$$



- ▶ 如果 $\liminf_{n \rightarrow \infty} E(X_n) = +\infty$, 则不等式显然成立. 以下设 $\liminf_{n \rightarrow \infty} E(X_n)$ 有限.
- ▶ 令 $Y_n = \inf_{k \geq n} (X_k - \underline{Y})$, $n \geq 1$, 则 $\{Y_n\}$ 为非负随机变量的上升序列, 且以 $\liminf_{n \rightarrow \infty} (X_n - \underline{Y})$ 为极限.
- ▶ 由单调收敛定理可得:

$$E \left[\liminf_{n \rightarrow \infty} (X_n - \underline{Y}) \right] = \lim_{n \rightarrow \infty} E \left[\inf_{k \geq n} (X_k - \underline{Y}) \right] \leq \liminf_{n \rightarrow \infty} E(X_n - \underline{Y}).$$

- ▶ 消去有限的 $E(\underline{Y})$ 可证得结论
- ▶ 对 $\{-X_n\}$ 用下确界的结论可证得上确界的结论.



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- ▶ 对 $\{-X_n\}$ 用下确界的结论可证得上确界的结论.



定理 0.12 (控制收敛定理) 设 $\{X_n\}$ 是一随机变量序列。如果存在可积随机变量 Y 使得 $|X_n| \leq Y$, 且 $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$, 则随机变量 $X(\omega)$ 可积, 且有

$$\int_{\Omega} \lim_{n \rightarrow \infty} X_n(\omega) P(d\omega) = \lim_{n \rightarrow \infty} \int_{\Omega} X_n(\omega) P(d\omega),$$

i.e. $E(X) = E(\lim_{n \rightarrow \infty} X_n) = \lim_{n \rightarrow \infty} E(X_n).$



► $|X_n| \leq Y$ 蕴含 $|X| \leq Y$, 故 X 可积.

► 取 $\underline{Y} = -Y$ 与 $\bar{Y} = Y$, 用 Fatou 引理便得到

$$E(X) \leq \liminf_{n \rightarrow \infty} E(X_n) \leq \limsup_{n \rightarrow \infty} E(X_n) \leq E(X)$$

► 故 $\lim_n E(X_n)$ 存在且等于 $E(X)$, 定理至此得证.



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定义 0.4 设 $(\Omega, \mathcal{F}, P)$ 是一个概率空间, 称随机变量 X 平方可积, 如果

$$E(X^2) < \infty.$$

注 0.1

- ▶ $(\Omega, \mathcal{F}, P)$ 上的所有平方可积随机变量构成的集合记为 $L^2(\Omega, \mathcal{F}, P)$ 或简记为 L^2 .
- ▶ 类似的, 对 $p \geq 1$, 也可以定义 L^p 空间, 即所有满足 $E(|X|^p) < \infty$ 的随机变量组成的集合.
- ▶ 若 $X \in L^2$, 则由 $E|X| < E(1 + X^2) < \infty$ 知 X 可积.



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定理 0.13 对任意的随机变量 X 与 Y 都有

$$[E(XY)]^2 \leq E(X^2)E(Y^2)$$

等式成立当且仅当存在常数 t_0 使得

$$P(Y = t_0 X) = 1$$

Cauchy-Schwarz 不等式的证明: $[E(XY)]^2 \leq E(X^2)E(Y^2)$



证明: 对任意的实数 t , 定义

$$u(t) := E(tX - Y)^2 = t^2E(X^2) - 2tE(XY) + E(Y^2).$$

显然, 对一切的 $t \in R$, $u(t) \geq 0$, 故方程 $u(t) = 0$ 或者没有实根或者有重根, 从而

$$\Delta = [2E(XY)]^2 - 4E(X^2)E(Y^2) \leq 0$$

即

$$[E(XY)]^2 \leq E(X^2)E(Y^2)$$

上述不等式等号成立, 当且仅当 $\Delta = 0$ 即 $u(t) = 0$ 存在一个重根 t_0 , 这时

$$u(t_0) = E(t_0X - Y)^2 = 0$$

从而由可知

$$P(t_0X - Y = 0) = 1 \Leftrightarrow P(Y = t_0X) = 1.$$

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命题 0.1 平方可积随机变量集 L^2 是一个线性空间.

证明: 假设 $X, Y \in L^2$, 则对任何 $\alpha, \beta \in \mathbb{R}$, 由柯西 - 施瓦茨不等式有

$$\begin{aligned} E[(\alpha X + \beta Y)^2] &= \alpha^2 E(X^2) + 2\alpha\beta E(XY) + \beta^2 E(Y^2) \\ &\leq \alpha^2 E(X^2) + 2|\alpha\beta| [E(X^2) E(Y^2)]^{1/2} + \beta^2 E(Y^2) \\ &< \infty \end{aligned}$$

所以 L^2 是一个线性空间.



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所以 L^2 是一个线性空间.

赫尔德 (Hölder) 不等式



定理 0.14 设 $p, q > 1$ 满足 $1/p + 1/q = 1$. 若 $X \in L^p$ 和 $Y \in L^q$, 则

$$XY \in L^1, \text{ 且 } E(|XY|) \leq [E(|X|^p)]^{1/p} [E(|Y|^q)]^{1/q}.$$

证明:

- ▶ 若 $E(|XY|) = 0$, 欲证不等式显然成立.
- ▶ 若 $E(|XY|) > 0$, 则必有 $E(|X|) > 0$ 和 $E(|Y|) > 0$.
- ▶ 如若不然, 则 $P(X=0)=1$ 或 $P(Y=0)=1$, 从而 $P(XY=0)=1$, 与 $E(|XY|) > 0$ 矛盾.

$$\begin{aligned} & \text{考虑非负函数的问题, 故可设 } X \geq 0, Y \geq 0. \\ & \text{因此, 对任意 } \lambda > 0 \text{ 有 } |XY| = \lambda |XY| \leq \lambda |X|^p + \lambda^{-1} |Y|^q. \\ & \text{取 } \lambda = [E(|X|^p)]^{1/p} [E(|Y|^q)]^{1/q} \text{ 代入上式, 得} \\ & E(|XY|) \leq [E(|X|^p)]^{1/p} [E(|Y|^q)]^{1/q} + \frac{1}{[E(|X|^p)]^{1/p} [E(|Y|^q)]^{1/q}} E(|Y|^q) \\ & \quad = [E(|X|^p)]^{1/p} [E(|Y|^q)]^{1/q} + [E(|X|^p)]^{1/p} [E(|Y|^q)]^{1/q} = 2[E(|X|^p)]^{1/p} [E(|Y|^q)]^{1/q}. \end{aligned}$$

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- ▶ 根据对数函数的凹性, 对 $x, y > 0$ 有: $\frac{1}{p} \ln x + \frac{1}{q} \ln y \leq \ln \left(\frac{x}{p} + \frac{y}{q} \right)$.
- ▶ 因此, 对任何 $x, y \geq 0$ 有: $x^{1/p} y^{1/q} \leq \frac{x}{p} + \frac{y}{q}$.
- ▶ 在上式中令 $x = |X|^p / E(|X|^p)$ 和 $y = |Y|^q / E(|Y|^q)$, 并对所得不等式取期望即得结论.

赫尔德 (Hölder) 不等式



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赫尔德 (Hölder) 不等式



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- ▶ 根据对数函数的凹性, 对 $x, y > 0$ 有: $\frac{1}{p} \ln x + \frac{1}{q} \ln y \leq \ln \left(\frac{x}{p} + \frac{y}{q} \right)$.
- ▶ 因此, 对任何 $x, y \geq 0$ 有: $x^{1/p} y^{1/q} \leq \frac{x}{p} + \frac{y}{q}$.
- ▶ 在上式中令 $x = |X|^p / E(|X|^p)$ 和 $y = |Y|^q / E(|Y|^q)$, 并对所得不等式取期望即得结论.

赫尔德 (Hölder) 不等式



定理 0.14 设 $p, q > 1$ 满足 $1/p + 1/q = 1$. 若 $X \in L^p$ 和 $Y \in L^q$, 则

$$XY \in L^1, \text{ 且 } E(|XY|) \leq [E(|X|^p)]^{1/p} [E(|Y|^q)]^{1/q}.$$

证明:

- ▶ 若 $E(|XY|) = 0$, 欲证不等式显然成立.
- ▶ 若 $E(|XY|) > 0$, 则必有 $E(|X|) > 0$ 和 $E(|Y|) > 0$.
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命题 0.2 对任何随机变量 X , 函数 $r \mapsto [E(|X|^r)]^{1/r}$ 在 $(0, \infty)$ 上不降。

证明:

► 取 $r > q > 0$ 并注意

$$\frac{1}{\frac{r}{q}} + \frac{1}{\frac{r}{r-q}} = q/r + (r-q)/r = 1.$$

► 对 $X = |X|^q$ 和 $Y = 1$ 应用上面定理即知

$$E(|X|^q) = E(|X|^q Y) \leq \left[E\left(|X|^q\right)^{\frac{r}{q}} \right]^{q/r} \left[E\left(1\right)^{\frac{r}{r-q}} \right]^{(r-q)/r} = [E(|X|^r)]^{q/r}$$

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定义 0.5 若 $X \in L^2$, 则称

$$D(X) := E[(X - EX)^2]$$

为随机变量 X 的方差, 有时我们也用 $Var(X)$ 来表示 X 的方差. 称方差 $D(X)$ 的正平方根 $\sqrt{D(X)}$ 为随机变量 X 的标准差, 记为 $\sigma(X)$ 或 σ_X .

定理 0.15 假设随机变量 X 的方差存在, 则

$$\begin{aligned} D(X) &= E[(X - EX)^2] = \int_{-\infty}^{+\infty} (x - EX)^2 dF(x) \\ &= \begin{cases} \sum_i (x_i - EX)^2 [F(x_i) - F(x_{i-1})] = \sum_i (x_i - EX)^2 P(X = x_i) \\ \int_{-\infty}^{+\infty} (x - EX)^2 p(x) dx \end{cases} \end{aligned}$$



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1. $D(X) = E[(X - EX)^2] \geq 0$, $D(X) = 0$ 当且仅当 $P(X = EX) = 1$;
2. 常数的方差为 0 即 $D(c) = 0$, 其中 c 为常数;
3. $D(X) = E(X^2) - (EX)^2$: 事实上, 根据方差的定义及期望的线性性质, 我们有

$$\begin{aligned} D(X) &= E[(X - EX)^2] = E[X^2 - 2X \cdot EX + (EX)^2] \\ &= E(X^2) - 2EX \cdot EX + (EX)^2 \\ &= E(X^2) - (EX)^2 \end{aligned}$$

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4. 若 $E(X^2) = 0$, 则 $E(X) = 0, D(X) = 0$. 事实上

$$0 \leq D(X) = E(X^2) - (EX)^2 = -(EX)^2 \leq 0$$

5. 若 a, b 为常数, 则 $D(aX+b) = a^2 D(X)$, 事实上,

$$\begin{aligned} D(aX+b) &= E(aX+b - E(aX+b))^2 = E(aX - aEX)^2 \\ &= E[a^2(X-EX)^2] = a^2 E[(X-EX)^2] = a^2 D(X) \end{aligned}$$

6. $E(a) = E(X-a)$. 并且恒有 $E(X-a) \geq 0$, 对取到最小值 $a = EX$ 时

$$E(a) = E(X-EX+EX-a) = E(X-EX)^2 + (EX-a)^2$$

证得



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6. $E(a) = E(X - EX) = 0$ 且恒有 $E(X - EX) = 0$ 时取到最小值 0.

$$E(a) = E(X - EX + EX - EX) = E(X - EX)^2 + (EX - EX)^2$$

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6. $f(c) = E(X - c)^2$ 当且仅当 $c = E(X)$ 时取到最小值:

$$f(c) = E(X - c)^2 = E[(X - EX) + (EX - c)]^2 = E(X - EX)^2 + (EX - c)^2 = D(X) + (EX - c)^2$$

证毕



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6. $f(c) := E(X - c)^2$ 当且仅当 $c = E(X)$ 时取到最小值: 由

$$f(c) = E(X - EX + EX - c)^2 = E(X - EX)^2 + [EX - c]^2$$

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定理 0.16 设随机变量 X 的期望与方差均存在, 则对任意的 $\epsilon > 0$ 有

$$P(|X - EX| \geq \epsilon) \leq \frac{D(X)}{\epsilon^2}$$

证明: 记 $a = EX$, 则

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计算二项分布期望的另一方法



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$$X = X_1 + X_2 + \cdots + X_n$$

其中 $X_k, k = 1, \cdots, n$ 为第 k 次伯努利试验成功的次数，显然服从两点分布。故

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再由期望的线性性知

$$E(X) = E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n) = np$$



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$$\begin{aligned} E(X^2) &= \sum_{k=0}^n k^2 P(X=k) = \sum_{k=1}^n k(k-1+1) C_n^k p^k (1-p)^{n-k} \\ &= \sum_{k=1}^n k(k-1) C_n^k p^k (1-p)^{n-k} + \sum_{k=1}^n k C_n^k p^k (1-p)^{n-k} \\ &= \sum_{k=2}^n k(k-1) C_n^k p^k (1-p)^{n-k} + np \\ &= n(n-1)p^2 \sum_{k=2}^n C_{n-2}^{k-2} p^{k-2} (1-p)^{(n-2)-(k-2)} + np \\ &= n(n-1)p^2 + np \end{aligned}$$

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泊松分布的期望与方差: $P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}$



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$$D(X) = E(X^2) - (EX)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

几何分布的期望与方差: $P(X = k) = (1 - p)^{k-1}p$



$$\begin{aligned}E(X) &= \sum_{k=1}^{\infty} k(1-p)^{k-1}p = p \sum_{k=1}^{\infty} \sum_{n=1}^k (1-p)^{k-1} = p \sum_{n=1}^{\infty} \sum_{k=n}^{\infty} (1-p)^{k-1} \\&= p \sum_{n=1}^{\infty} (1-p)^{n-1} \frac{1}{1-(1-p)} = \sum_{n=1}^{\infty} (1-p)^{n-1} = \frac{1}{p} \\E(X^2) &= \sum_{k=1}^{\infty} k^2(1-p)^{k-1}p = p \left[\sum_{k=1}^{\infty} k(k+1)(1-p)^{k-1} - \sum_{k=1}^{\infty} k(1-p)^{k-1} \right] \\&= 2p \sum_{k=1}^{\infty} \sum_{n=1}^k n(1-p)^{k-1} - p \sum_{k=1}^{\infty} k(1-p)^{k-1} \\&= 2p \sum_{n=1}^{\infty} n \sum_{k=n}^{\infty} (1-p)^{k-1} - \frac{1}{p} = 2p \sum_{n=1}^{\infty} n(1-p)^{n-1}/p - \frac{1}{p} \\&= \frac{2}{p^2} - \frac{1}{p} \\D(X) &= E(X^2) - (EX)^2 = \frac{2}{p^2} - \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}\end{aligned}$$

几何分布的期望与方差: $P(X=k) = (1-p)^{k-1}p$



$$\begin{aligned}E(X) &= \sum_{k=1}^{\infty} k(1-p)^{k-1}p = p \sum_{k=1}^{\infty} \sum_{n=1}^k (1-p)^{k-1} = p \sum_{n=1}^{\infty} \sum_{k=n}^{\infty} (1-p)^{k-1} \\&= p \sum_{n=1}^{\infty} (1-p)^{n-1} \frac{1}{1-(1-p)} = \sum_{n=1}^{\infty} (1-p)^{n-1} = \frac{1}{p} \\E(X^2) &= \sum_{k=1}^{\infty} k^2(1-p)^{k-1}p = p \left[\sum_{k=1}^{\infty} k(k+1)(1-p)^{k-1} - \sum_{k=1}^{\infty} k(1-p)^{k-1} \right] \\&= 2p \sum_{k=1}^{\infty} \sum_{n=1}^k n(1-p)^{k-1} - p \sum_{k=1}^{\infty} k(1-p)^{k-1} \\&= 2p \sum_{n=1}^{\infty} n \sum_{k=n}^{\infty} (1-p)^{k-1} - \frac{1}{p} = 2p \sum_{n=1}^{\infty} n(1-p)^{n-1}/p - \frac{1}{p} \\&= \frac{2}{p^2} - \frac{1}{p} \\D(X) &= E(X^2) - (EX)^2 = \frac{2}{p^2} - \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}\end{aligned}$$

几何分布的期望与方差: $P(X=k) = (1-p)^{k-1}p$



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几何分布的期望与方差: $P(X=k) = (1-p)^{k-1}p$



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$$\begin{aligned} E(X^2) &= \sum_{k=1}^{\infty} k^2(1-p)^{k-1}p = p \left[\sum_{k=1}^{\infty} k(k+1)(1-p)^{k-1} - \sum_{k=1}^{\infty} k(1-p)^{k-1} \right] \\ &= 2p \sum_{k=1}^{\infty} \sum_{n=1}^k n(1-p)^{k-1} - p \sum_{k=1}^{\infty} k(1-p)^{k-1} \\ &= 2p \sum_{n=1}^{\infty} n \sum_{k=n}^{\infty} (1-p)^{k-1} - \frac{1}{p} = 2p \sum_{n=1}^{\infty} n(1-p)^{n-1}/p - \frac{1}{p} \\ &= \frac{2}{p^2} - \frac{1}{p} \end{aligned}$$

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负二项分布的期望与方差: $P(X=k) = C_{k-1}^{r-1} p^r (1-p)^{k-r}$



$$\begin{aligned} E(X) &= \sum_{k=r}^{\infty} k C_{k-1}^{r-1} p^r (1-p)^{k-r} = \sum_{k=r}^{\infty} \frac{r}{p} \frac{k \cdot (k-1)!}{r \cdot (r-1)! (k-r)!} p^{r+1} (1-p)^{k-r} \\ &= \sum_{k=r}^{\infty} \frac{r}{p} C_{k+1-1}^{r+1-1} p^{r+1} (1-p)^{k-r} = \frac{r}{p} \sum_{l=r+1}^{\infty} C_{l-1}^{r+1-1} p^{r+1} (1-p)^{l-(r+1)} = \frac{r}{p} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \sum_{k=r}^{\infty} k^2 C_{k-1}^{r-1} p^r (1-p)^{k-r} = \sum_{k=r}^{\infty} k(k+1-1) C_{k-1}^{r-1} p^r (1-p)^{k-r} \\ &= \sum_{k=r}^{\infty} k(k+1) C_{k-1}^{r-1} p^r (1-p)^{k-r} - \sum_{k=r}^{\infty} k C_{k-1}^{r-1} p^r (1-p)^{k-r} \\ &= \sum_{k=r}^{\infty} \frac{r(r+1)}{p^2} C_{k+2-1}^{r+2-1} p^{r+2} (1-p)^{(k+2)-(r+2)} - \frac{r}{p} \\ &= \frac{r(r+1)}{p^2} - \frac{r}{p} \end{aligned}$$

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负二项分布的期望与方差: $P(X=k) = C_{k-1}^{r-1} p^r (1-p)^{k-r}$



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$$\begin{aligned} E(X^2) &= \sum_{k=r}^{\infty} k^2 C_{k-1}^{r-1} p^r (1-p)^{k-r} = \sum_{k=r}^{\infty} k(k+1-1) C_{k-1}^{r-1} p^r (1-p)^{k-r} \\ &= \sum_{k=r}^{\infty} k(k+1) C_{k-1}^{r-1} p^r (1-p)^{k-r} - \sum_{k=r}^{\infty} k C_{k-1}^{r-1} p^r (1-p)^{k-r} \\ &= \sum_{k=r}^{\infty} \frac{r(r+1)}{p^2} C_{k+2-1}^{r+2-1} p^{r+2} (1-p)^{(k+2)-(r+2)} - \frac{r}{p} \\ &= \frac{r(r+1)}{p^2} - \frac{r}{p} \end{aligned}$$

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标准正态分布的期望与方差: $X \sim N(0, 1)$



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$$E(X) = \int_{-\infty}^{+\infty} xp(x)dx = \int_{-\infty}^{+\infty} x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = 0$$

$$\begin{aligned} D(X) &= E[(X - EX)^2] = E(X^2) = \int_{-\infty}^{+\infty} x^2 p(x) dx \\ &= \int_{-\infty}^{+\infty} x^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x d(-e^{-\frac{x^2}{2}}) \\ &= \frac{1}{\sqrt{2\pi}} \left(-xe^{-\frac{x^2}{2}} \Big|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} dx \right) \\ &= 1 \end{aligned}$$

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